

8 .Database Triggers

Implement and test triggers to maintain data integrity in the database.

```
mysql> create database trigger_db;  
Query OK, 1 row affected (0.13 sec)
```

```
mysql> use trigger_db;  
Database changed  
mysql> create table emp(empid int(20)Primary key,empname varchar(50),salary int(30));  
Query OK, 0 rows affected, 2 warnings (1.15 sec)
```

```
mysql> insert into emp values(1,"Ram",50000),(2,"Gita",60000),(3,"Sham",55000);  
Query OK, 3 rows affected (0.11 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> select * from emp;  
+-----+-----+-----+  
| empid | empname | salary |  
+-----+-----+-----+  
| 1 | Ram | 50000 |  
| 2 | Gita | 60000 |  
| 3 | Sham | 55000 |  
+-----+-----+-----+  
3 rows in set (0.00 sec)
```

1. Trigger to Maintain Data Integrity:

```
mysql> drop trigger if exists chk_salary;  
-> delimiter //  
Query OK, 0 rows affected, 1 warning (0.04 sec)  
mysql> create trigger chk_salary  
-> before insert on emp  
->  
-> for each row  
-> begin  
-> if NEW.salary <= 0 then  
-> signal sqlstate '45000'  
-> set message_text="Salary can not be negative and zero!";  
-> end if;  
-> end;  
-> //  
Query OK, 0 rows affected (0.09 sec)
```

```
mysql> delimiter ;
```

```
mysql> insert into emp values(4,"Sita",0);  
ERROR 1644 (45000): Salary can not be negative and zero!
```

2.Trigger to Prevent Deleting Employees with High Salary

/*Business Rule:Employees earning more than 50,000 cannot be deleted.*/

```
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER high_salary
-> BEFORE DELETE ON emp
-> FOR EACH ROW
-> BEGIN
->   IF OLD.salary > 50000 THEN
->     SIGNAL SQLSTATE '45000'
->     SET MESSAGE_TEXT = 'Cannot delete employee with salary greater than 50000';
->   END IF;
-> END;
-> //
```

Query OK, 0 rows affected (0.15 sec)

```
mysql>
mysql> DELIMITER ;
```

```
mysql> delete from emp where salary=60000;
ERROR 1644 (45000): Cannot delete employee with salary greater than 50000
```

3.Trigger to Automatically Log Employee Inserts (Audit Integrity)

```
mysql> DROP TRIGGER IF EXISTS emp_insert;
Query OK, 0 rows affected, 1 warning (0.06 sec)
```

```
mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER emp_insert
-> AFTER INSERT ON emp
-> FOR EACH ROW
-> BEGIN
->   INSERT INTO emp2 (empid, action, action_date)
->   VALUES (NEW.empid, 'INSERT', NOW());
-> END;
-> //
Query OK, 0 rows affected (0.16 sec)
```

```
mysql>
mysql> DELIMITER ;
```

```
mysql> insert into emp values(4,"Radha",50000);
Query OK, 1 row affected (0.09 sec)
```

```
mysql> select * from emp;
+-----+-----+-----+
| empid | empname | salary |
+-----+-----+-----+
| 2 | Gita | 60000 |
| 3 | Sham | 55000 |
| 4 | Radha | 50000 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> select * from emp2;
+-----+-----+-----+
| empid | action | action_date |
+-----+-----+-----+
| 4 | INSERT | 2026-04-22 13:57:01 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

4. Trigger to Automatically Set Default Salary

```
mysql> CREATE TRIGGER default_salary
-> BEFORE INSERT ON emp
-> FOR EACH ROW
-> BEGIN
->   IF NEW.salary IS NULL THEN
->     SET NEW.salary = 30000;
->   END IF;
-> END;
-> //
```

Query OK, 0 rows affected (0.18 sec)

```
mysql>
mysql> DELIMITER ;
```

```
mysql> insert into emp values(7,"Raj",NULL);
Query OK, 1 row affected (0.08 sec)
```

```
mysql> select * from emp;
```

```
+-----+-----+-----+
| empid | empname | salary |
+-----+-----+-----+
| 2 | Gita | 60000 |
| 3 | Sham | 55000 |
| 7 | Raj | 30000 |
+-----+-----+-----+
```

3 rows in set (0.00 sec)